Solution to the Daily Limit

Based on the problem provided in the file dailyintegral-limit-limit-1x1-2026-08-20.jpg, we are tasked with evaluating the following limit involving a sum S_n over points on a circle:

$$\lim_{n\to\infty} n^2 \left(\frac{\pi^2}{3} - S_n \right)$$

where n points are equally spaced on a circle of radius $R = \frac{n}{2\pi}$, and S_n is the sum of the reciprocals of the squares of the straight-line distances from one fixed point to all the other $n-1$ points.

1. Formula for the Distances

Let the fixed point be located at an angle of 0 on the circle. Since the n points are equally spaced, their angular positions are given by $\theta_k = \frac{2\pi k}{n}$ for $k = 1, 2, \ldots, n-1$.

The straight-line distance d_k between the point at $\theta = 0$ and the point at θ_k can be found using the law of cosines or the standard chord length formula:

$$d_k = 2R \sin\left(\frac{\theta_k}{2}\right)$$

Substituting $R = \frac{n}{2\pi}$ and $\theta_k = \frac{2\pi k}{n}$, we get:

$$d_k = 2\left(\frac{n}{2\pi}\right) \sin\left(\frac{\pi k}{n}\right) = \frac{n}{\pi} \sin\left(\frac{\pi k}{n}\right)$$

2. Evaluating the Sum S_n

The sum S_n of the reciprocals of the squares of these distances is:

$$\begin{aligned} S_n &= \sum_{k=1}^{n-1} \frac{1}{d_k^2} \\ &= \sum_{k=1}^{n-1} \frac{1}{\left(\frac{n}{\pi} \sin\left(\frac{\pi k}{n}\right)\right)^2} \\ &= \frac{\pi^2}{n^2} \sum_{k=1}^{n-1} \csc^2\left(\frac{\pi k}{n}\right) \end{aligned}$$

We can use the well-known trigonometric identity for the sum of squared cosecants of equally spaced angles:

$$\sum_{k=1}^{n-1} \csc^2\left(\frac{\pi k}{n}\right) = \frac{n^2-1}{3}$$

Substituting this identity back into our expression for S_n:

$$S_n = \frac{\pi^2}{n^2}\left(\frac{n^2-1}{3}\right) = \frac{\pi^2}{3} - \frac{\pi^2}{3n^2}$$

3. Evaluating the Limit

We are now ready to evaluate the limit:

$$L = \lim_{n\to\infty} n^2 \left(\frac{\pi^2}{3} - S_n \right)$$

Substitute the simplified expression for S_n into the limit:

$$\begin{aligned} L &= \lim_{n\to\infty} n^2 \left[\frac{\pi^2}{3} - \left(\frac{\pi^2}{3} - \frac{\pi^2}{3n^2} \right) \right] \\ &= \lim_{n\to\infty} n^2 \left(\frac{\pi^2}{3n^2} \right) \\ &= \lim_{n\to\infty} \frac{\pi^2}{3} \end{aligned}$$

Since the variables exactly cancel out and the expression inside the limit is a constant, evaluating it as $n \to \infty$ is straightforward.

Conclusion

Therefore, the evaluated limit is:

$$\lim_{n\to\infty} n^2 \left(\frac{\pi^2}{3} - S_n \right) = \frac{\pi^2}{3}$$